第十三周作业.

17.12. (1) $\begin{cases} d\sin\theta_1 = k\lambda. \\ d\sin\theta_2 = (k+1)\lambda. \end{cases}$ 则 $d = \frac{600nm}{0.3-0.2} = 6\mu m$

(2) $\begin{cases} a\sin\theta = k_1\lambda \\ d\sin\theta = k_2\lambda. \end{cases}$ 则 $\frac{a}{d} = \frac{k_1}{4}$.

$k_1 = 1$ 时. $a_{min} = \frac{1}{4} \times 6\mu m = 1.5\mu m$.

(3). $d\sin\theta = k\lambda$.

$k < \frac{d}{\lambda} = 10$. 则 $k = 0, \pm1, \pm2, \ldots, \pm9$.

缺级: $k_2 = \frac{d}{a}k_1 = 4k_1$, $k = \pm1, \pm2$.

则 $k_2 = \pm4, \pm8$.

则实际共15级, 分别为 $0, \pm1, \pm2, \pm3, \pm5, \pm6, \pm7, \pm9$ 级.

17.17. $d\sin\theta = k\lambda$.

$d = \frac{2 \times 600nm}{\frac{1}{2}} = 2.4\mu m$.

$R = kN = \frac{\lambda}{\Delta\lambda}$.

$2N = \frac{600nm}{0.05nm}$, 则 $N = 6000$.

$2.4\mu m \times \frac{1}{2} = k \cdot \lambda$.

则 $\lambda = 400nm$, $k = 3$.

$\begin{cases} a\sin\theta = k'\lambda. \\ d\sin\theta = k\lambda. \end{cases}$ $k = k' \cdot \frac{d}{a} = 3$. 则 $k' = 1$, $\begin{cases} a = 0.8\mu m. \\ b = 1.6\mu m \end{cases}$

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20.2. $T_1\lambda_{m1} = T_2\lambda_{m2}$.

$$\frac{T_2}{T} = \frac{\lambda_{m1}}{\lambda_{m2}} = \frac{69}{50}$$

$$M_B(T) = \sigma T^4$$

$$\frac{M_B(T)_2}{M_B(T)_1} = \frac{T_2^4}{T_1^4} = 3.63$$

20.5. $(4\pi R_{太}^2) \times \sigma T^4 = (4\pi d^2) \times M$.

$$M = \frac{R_{太}^2}{d^2} \times \sigma T^4$$

$$M \times \pi R_{地}^2 = M_{地} \cdot 4\pi R_{地}^2 = \sigma T_{地}^4 \cdot 4\pi R_{地}^2$$

$$\text{则 } T_{地} = \sqrt[4]{\frac{M}{4\sigma}} = \sqrt[4]{\frac{R_{太}^2}{4d^2} \times T^4} = 280.3K.$$

20.7. (1) $E_{max} = h\nu - A = 6.63\times10^{-34} J\cdot s \times \frac{3\times10^8 m/s}{200nm} - 4.2eV$

$E_{min} = 0$. $= 2.02eV$.

(2) $U = \frac{E_{max}}{e} = 2.02V$.

(3). $h\nu = A$.

$$\lambda = \frac{c}{\nu} = \frac{ch}{A} = \frac{3\times10^8 m/s \times 6.63\times10^{-34} J\cdot s}{4.2eV}$$

$$= 296 nm.$$

20.16 $E_k = h\Delta\nu$ $c = \lambda\nu$. $\lambda' = 1.2\lambda$, $\nu' = \frac{5}{6}\nu$.

$$E_k = h\nu - h\times\frac{5}{6}\nu = 0.6MeV \times \frac{1}{6}\nu = 0.1 MeV.$$

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20.17 $m=\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}=1.25m_0$.

$\Delta E=\Delta mc^2=0.25m_0\times c^2$.

$\Delta E=h\Delta\nu=h(\frac{c}{\lambda_1}-\frac{c}{\lambda_2})=0.25m_0c^2$

解得 $\lambda_2=0.0043nm$

$\Delta\lambda=\frac{h}{m_0c}(1-\cos\varphi)$, $\cos\varphi=1-\frac{0.0043nm-0.003nm}{0.0024nm}=\frac{11}{24}$

$\varphi=\arccos\varphi=62.7^\circ$

21.1(1) $E=100V\cdot e=\frac{1}{2}mv^2$. $v=\sqrt{\frac{2\times100\times1.6\times10^{-19}C}{m}}$

$\lambda=\frac{h}{m_0v}=1.23\times10^{-10}m$

$=0.123nm$.

(2). $\lambda=\frac{h}{mv}$

$=\frac{6.63\times10^{-34}J\cdot s}{10^{-2}kg\times800m/s}=8.29\times10^{-35}m$.

21.2 解: $\lambda=\frac{h}{mv}$.

$v=\frac{h}{m\lambda}=\frac{6.63\times10^{-34}J\cdot s}{9.1\times10^{-31}kg\times0.2nm}=3.64\times10^{6}m/s$

$P_电=mv=\frac{h}{\lambda}=\frac{6.63\times10^{-34}J\cdot s}{0.2nm}=3.315\times10^{-24}kg\cdot m/s$.

$E=\sqrt{m_0^2c^4+P_电^2c^2}$

$=5.12\times10^{5}eV$.

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解：$p = \frac{h}{\lambda} = \frac{6.63\times10^{-34} J\cdot s}{0.2\,nm} = 3.315\times10^{-24} kg\cdot m/s.$

$E = h\nu$

$= 6.63\times10^{-34} J/s \cdot \frac{3\times10^{8} m/s}{0.2\,nm}$

$= 6.2\times10^{3} eV.$

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